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ctive 1.

W. A. P. in java to print hello world.

```
Public class new {
    Public static void main (string [] args)
    {
        System. Out . Println (" Hello world");
    }
}
```

output :- Hello world

ctive - 2

W. A. P. in java to calculate the sum of 2 number.

```
Public class new {
    Public static void main (string [] args)
    {
        int a = 10;
        int b = 20;
        int sum = a+b ;
        System. Out. Println ("The sum of a and b
                                is : " + sum);
    }
}
```

Output :-

Sum = 30

ctive 3.

W. A. P in Java to find the greatest of three number.

```
Public class greatest {
    Public static void main (String [] args)
    {
        int a = 10;
        int b = 20;
        int c = 50;
      if ( a > b && a > c ) {
          System. Out. println ("a is greatest");
      }
      else if ( b > a && b > c ) {
          System. Out. println (" b is greatest");
      } else
      {
          System. Out. println ("c is the greatest");
      }
    }
}
```

output.

c is the greatest

ective 5

W.A.P to sort an array of 10 number.

```
Public class Array {
    Public static void main (String [] args)
    {
        int [] arr = {5, 2, 8, 1, 3, 4, 6, 7, 9, 10};
        Sort Array (arr);
        for (int num: arr) {
            System. Out. println (num + " ");
        }
    }

    Public static void Sort Array (int [] arr)
    {
    for (int i=0 ; i< arr.length-1 ; i++) {
    for (int j=0 ; j< arr. length - i-1 ; j++)
    {
        if (arr [j] > arr [j+1]) {
            int temp = arr [j];
            arr [j] = arr [j+1];
            arr [j+1] = temp ;
        }
    }
    }
}
```

jective .6.

W.A.P to implement class and object in Java.

```
Public class student {
    String name;
    int age;
    Void display () {
        System. Out. println ("Name:" + name);
        System. Out. println (" Age:" + age);
    }

    Public static void main (String [] args) {
        student S1 = new student ();
        S1. name = "Saurabh";
        S1.. age = 20;
        S1. display ();
    }
}
```

Output

Name : Saurabh

age : 20

Objective 7 :- Write a java program to store 6 country names in array and sort them.

```
import java.util.Arrays;
Public class sort countries {
    Public static void main (String [] args) {
String[] countries = {"India", "USA", "Japan", "Brazil",
        "Canada", "Australia"};

System.out.Println("Before sorting:");
    for (string c: countries)
        System.out.Println(c);

    Array.sort(countries);
    System.out.Println("After");
        for (String c: countries)
            System.out.Println(c);

    }

}
```

Output.

Before sorting

India, USA, Japan, Brazil, Canada, Australia.

After

Austrialia, Brazil, Canada, India, Japan, USA.

Objective 8 :- Write a Java Program which includes compare substring and demstrate string concatenation.

```
Public class Substring compare{
Public static void main (String [] args)
{
    String S1 = " Hello";
    String S2 = " hello";
    String sub1 = S1 . substring (0,5);
    String sub2 = S2 - substring (0,5);
if ( sub1 . equals (sub2))
        System . out . Println (" Substring are equal");
else
        system. out. Println ("Not equal!");
if ( sub1. equal Is Ignore Case (sub2))
        System. out . Println (" substring are equal (Case-insensitive)");
else
    System. out. Println ("substring are not equal") ;
    String concat = sub1 + sub2 ;
    System . out. Println (" concatenated lower case :
            + concatenated . to lower case() );
    System. out. Println (" concatenated upper case : " +
            concate.        upper case() );
}
}
```

Output

Objective :9 :- W.A.P to implement constructor.

```
class J {
        String name ;
          int age ;
        String number ;

    J() {
        System.out.Println ("welcome");
    }
    J (String n , int a ; String nu ) {
            name = n ;
            age = a ;
            number = nu ;
        System.out.Println ("name " + name + "age" + age
            + nu ) ;
    }
    Public static void main (String args [] ) {
        J obj = new J () ;
        J.obj = new J ("Virat", 36 , "8963534323) ;
    }
```

output

welcome

name Virat age 36. 8963534323 .

Objective - 10 :- write A program to implement method overloading

```
class O{
    void K (int a, int b){
        System. out. println ("sum is " + (a+b));
    }
    void K (double a, double b) {
        System. out. println ("sum is + " (a+b));
    }
    Public static void main (string args[]) {
        O obj = new O ();
        obj. K (2, 3);
        obj. K (2.3, 3);
    }
}
```

Output.

```
sum is 5
sum is 5.3
```

objective 4) W.A.P to implement overriding.

```
class A {
    void k ( string n, string id ) {
        System. out. println ("name "+n+" id "+id);
    }
}
class B extends A {
    void k (string n, string id ) {
        super .k (n, id);
        system. out. println ("Example of
            overriding
    }
}
class C {
    Public static void main (string .args [ ] ) {
        A obj.2 = new B ();
        obj2.k ("CSE"; "405");
    }
}
```

Output.

name CSE id 405

this Example of overriding.

Objective 4 W.A.P a java to calculate factorial of given number.

```
import java.util.scanner;
public class factorial {
    public static void main (string [] args){
        int n, fact =1;
    scanner sc = new scanner (system.in).
    system.out.println ("Enter a number");
        n = sc.next Int ();
    for (int i=1; i<=n; i++) {
        fact = fact * i;
    }
    system.out.println ("factorial =" + fact);
    }
}
```

Output

Enter a number 5

factorial = 120.

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Objective :-13 W.A.P to implement abstraction using abstract class.

```
abstract class B {
    abstract void run ();
}
class H extends B
{
    void run () {
    System.out.println ("running");
}
}
Public class A {
public static void main (string args[]) {
    B ob = new H();
    ob.run ();
}
}
```

output

running.

Objective .14. W.A.P to implement abstraction using interface.

```
interface Notepad {
    void write (string text);
    void display ();
}

class MyNotepad implements Notepad {
    String data = " ";
    public void write (String text)
    {
        data += text;
    }
    public void display () {
    System.out.println ("Content: " + data);
    }
    Public static void main (String args[]) {
    Notepad note = new MyNotepad ();
        note.write ("Hello");
        note.display ();
    }
}
```

Output

Hello.

Objective. 15 W.A.P to Import package.

```
Package my pack ;
Public class My {
Public void show () {
    System . out . println ("This is package");
  }
}
```

using package.

```
import my pack . my ;
Public class Test {
    Public static void main ( String [] args){
  my Obj = new my ();
        obj . show ();
  }
}
```

Output → This is package.

objective. 16. W.A.P for java to illustrate the use of Null pointer exception.

```
Public class N {
    Public static void main (String [] args) {
    String m= Null;
    try
     {
     system. out. Println ("This is exception" + m.length());
     }
    Catch (Null PointerException e)
     {
     system . out. Println ("Null Pointer error");
     }
  }
}
```

Output.

Null Pointer error.

objective. 18. W.A.P in java to illustrate the use of Array Index out of Bounds Exception.

```
Public class A{
    Public static Void main(String args[]){
    int a[] = { 1,2,3,4,5}
    try {
    System.out.Println("Exception" + a[6]);
    }
    Catch (Array Index out of Bounds Exception e){
    System.out.Println("Array Index out of
                Bounds Exception");
    }
  }
}
```

Output.

Array Index out of Bounds Exception.

Jective. 19 W.A.P in java to illustrate the use of of Arithmetic Exception.

```
Public class K {
    Public static void main (String args[]) {
    int s = 1;
    try {
    s=s/0;
    System.out.println ("Arithmatic error"+s);
    }
    catch (ArithmaticException e);
    {
    System.out.println ("Arithmatic Exception");
    }
    }
}
```

Output

Arithmatic Exception.

objective 20 W.A.P to implement multi threading in java

```
class C extends Threads
Public void run ()
{
System. out. Println("class A");
}
}
Class b extends Threads.
{
System . out. Printen ("class.b");
}
}

Class A {
Public static void main (string args[])
{
C a1 = new C();
a1. start ();
B b1 = new B();
b1. start ();
}
}
```

Output :

```
Class A
class b
```

Objective 21 : W.A.P to implement use of suspend, wait & sleep.

```
class Table {
  synchronized public void prt (int n) {
    for (int i = 1; i <= 10; i++) {
      System.out.Println(n+"x"+i+"="+(n*i));
    try {
      Thread.sleep (400); }
    Catch (InterruptedException e) {
      System.out.Println (e);
    }
  }
}
}

class C extends Thread {
  Table t;
  C(Table t) {
    this.t = t;
  } public void run () {
      t.prt(5); }}
class B extends Thread {
  Table t;
  B(Table t) {
    this.t = t; }
  Public void run () {
      t.prt (10); }}

class wait {
  Public static void main (String args[]) {
    Table obj = new Table ();
    C a1 = new C(obj);
    B b1 = new B(obj);
    a1.start ();
    b1.start ();
  }
}
```

objective 22. W.A.P to implement thread priority

```
class A extends Thread {
Public void run () {
  System. out. Println ("class A");
 }
}

Class B extends Thread{
  Public void run () {
  System out. println ("class B");
 }
}

class Priority {
  Public static void main (String args[]) {
  A a1 = new A();
  B b1 = new B();

  a1. set Priority (Thread. MIN_ PRIORITY);
  b1. set Priority (Thread. MAX_ PRIORITY);

   a1. start ();
   b1. Start ();
  }
}
```

Output

```
class A
class B
```

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